Semmai

Guaranteeing semantic intent preservation between Natural Language requests and generated SQL through Action-Graph alignment and Tamil NLP synthesis.

THE CRITICAL AI FLAW

The Text-to-SQL Hallucination Problem

Silent Data Distortion

LLMs generate syntactically correct SQL that silently alters the query intent—flipping operators, dropping constraints, or drifting values without throwing errors.

Boundary & Negation Flips

A simple request for silently transformed into or mapped to leads to catastrophic financial or operational misreports.

The Solution: Semmai proves mathematically whether generated SQL preserves the original user intent prior to database execution.

END-TO-END PIPELINE

Architecture & Verification Workflow



Hybrid Reasoning: Combines deterministic graph-matching algorithms with optional SBERT/TF-IDF neural embeddings for soft value alignment.

CORE ABSTRACTION

Universal Semantic Intent Graph (USIG)

The USIG represents user requests as canonical semantic structures independent of target SQL dialects, syntax, or phrasing variations.

- Entities & Primary Target Isolation
- Constraint & Canonicalized Value Triples
- Projections, Groupings & Aggregations



PROOF OBLIGATIONS

The Semantic Proof Checks

1. Entity Check

Validates that intended primary tables are queried without extraneous joins.

2. Constraint Check

Detects value drift, boundary shifts, and negation flips in WHERE clauses.

3. Aggregation

Verifies correct mapping of SUM, AVG, COUNT, MIN, and MAX functions.

4. Order & Limit

Ensures ranking column, direction (ASC/DESC), and size limits match.

Additional Checks: Logic Conjunction Preservation (AND vs OR), Column Projection, Grouping Equivalence, and Action Match (SELECT vs DELETE).